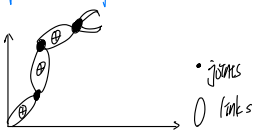
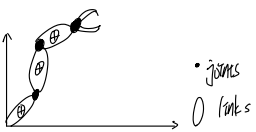
· Serial / open-chain systems (robot arms)

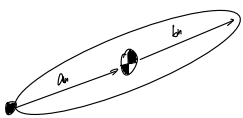


For now we assume there's no loop

Can easily compute the forward kinematics

Eg. 2D





$\theta_1 = q_1$

$^1\vec{r}_{1/0} = \begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) \end{bmatrix} a_1$

Link com position $R(\theta_1)$ — the vector from joint 1 to com of link 1 in joint frame

in world frame

for $n = 2:N$

$\theta_n = \theta_{n-1} + q_n$

$r_{n/0} = r_{n-1/0} + R(\theta_{n-1})\, b_{n-1} + R(\theta_n)\, a_n$

vector from com of link $n-1$ to joint n, in joint frame

Eg. 3D

$R_1 = S_1(q_1)$

$^1r_{1/0} = R_1 a_1$

$\begin{cases} R_n = R_{n-1} \cdot S_1(q_n) \\ ^1r_{n/0} = {}^1r_{n-1/0} + R_{n-1} b_{n-1} + R_n a_n \end{cases}$

Dynamics in 2D

$\theta_1 = q_1$

$r_{1/0} = \begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) \end{bmatrix} a_1$

for $n = 2:N$

$\theta_n = \theta_{n-1} + q_n$

$r_n = r_{n-1} + R_{n-1} b_{n-1} + R_n a_n$

$$\begin{bmatrix} r_1 \\ \theta_1 \\ r_2 \\ \theta_2 \\ \vdots \\ r_N \\ \theta_N \end{bmatrix} = k(q)$$

kinematic function

$$\frac{d}{dt}\begin{bmatrix} r_1 \\ \theta_1 \\ \vdots \\ r_N \\ \theta_N \end{bmatrix} = \frac{\partial k}{\partial q}\dot{q}$$

$\frac{d}{dt}\begin{bmatrix} r_1 \\ \theta_1 \\ \vdots \\ r_N \\ \theta_N \end{bmatrix}$: inertial velocity

$\frac{\partial k}{\partial q}$: kinematic Jacobian $3n \times n$

kinetic energy

$$T = \sum_{i=1}^{N} \frac{1}{2} m_i \|\dot{r}_i\|^2 + \frac{1}{2} J_i \dot{\theta}_i^2$$

$$= \frac{1}{2} \begin{bmatrix} \dot{r}_1 & \dot{\theta}_1 & \cdots & \dot{r}_N & \dot{\theta}_N \end{bmatrix} \underbrace{\begin{bmatrix} m_1 I & & & & \\ & J_1 & & & \\ & & \ddots & & \\ & & & m_N I & \\ & & & & J_N \end{bmatrix}}_{\bar{M}} \begin{bmatrix} \dot{r}_1 \\ \dot{\theta}_1 \\ \vdots \\ \dot{r}_N \\ \dot{\theta}_N \end{bmatrix}$$

$\bar{M}$: mass matrix in maximal coordinates

always constant, independent of q

$$= \frac{1}{2} \dot{q}^T \underbrace{\frac{\partial k}{\partial q}^T \bar{M} \frac{\partial k}{\partial q}}_{M(q)} \dot{q}$$

potential energy: $U(q) = U(k(q))$ similar to kinetic energy

Lagrangian: $L = T - U$

$$= \frac{1}{2} \dot{q}^T M(q) \dot{q} - U(q)$$

Euler-Lagrange: $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \underbrace{\tau}_{\text{joint torque}} + \underbrace{\sum_{i=1}^{N} Dr_i(q)^T F_i^{(ext)}}_{\text{other forces}}$

- Differentiating through quaternions

(use q for generalized coordinates, Q for quaternions)

$$\dot{Q}_L = Q_L * \begin{bmatrix} 0 \\ \frac{1}{2}\omega_B \end{bmatrix} = \frac{1}{2} L(Q_L) \begin{bmatrix} 0 \\ \omega_B \end{bmatrix} = \frac{1}{2} L(Q_L) H \omega_B \qquad H = \begin{bmatrix} 0 \\ I \end{bmatrix}$$

$$\delta Q = \begin{bmatrix} \cos(\frac{\theta}{2}) \\ \hat{\omega} \sin(\frac{\theta}{2}) \end{bmatrix} \approx \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{2}\hat{\omega}\theta \end{bmatrix}$$

$$Q * \delta Q \approx Q * \begin{bmatrix} 1 \\ 0 \end{bmatrix} + Q * \begin{bmatrix} 0 \\ \frac{1}{2}\hat{\omega}\theta \end{bmatrix}$$

① $f: Q \to \mathbb{R}^n$

$$f(Q * \delta Q) \approx f(Q + Q * \begin{bmatrix} 0 \\ \frac{1}{2}\hat{\omega}\theta \end{bmatrix})$$
$$= f(Q + L(Q) \begin{bmatrix} 0 \\ \frac{1}{2}\hat{\omega}\theta \end{bmatrix})$$
$$\approx f(Q) + \frac{\partial f}{\partial Q} \frac{1}{2} L(Q) H \cdot (\hat{\omega}\theta)$$

$G(Q)$: Attitude Jacobian, map from axis-angle to jacobian

$x = f(Q) \Rightarrow \delta x = \frac{1}{2} \frac{\partial f}{\partial Q} G(Q) (\hat{\omega}\theta)$

axis-angle → Quaternion $\xrightarrow{f} x$, this cares $\partial f / \partial (\hat{\omega}\theta)$

② $f: \mathbb{R}^n \to Q$

$$Q * \delta Q \approx Q + Q * \begin{bmatrix} 0 \\ \frac{1}{2}\hat{\omega}\theta \end{bmatrix}$$
$$\approx Q + L(Q) H \frac{1}{2} (\hat{\omega}\theta)$$

$$f(x + \delta x) \approx f(x) + \frac{\partial f}{\partial x} \delta x$$

$$\frac{1}{2} L(Q) H (\hat{\omega}\theta) = \frac{\partial f}{\partial x} \delta x$$
$$H (\hat{\omega}\theta) = 2 L(Q)^T \frac{\partial f}{\partial x} \delta x$$
$$(\hat{\omega}\theta) = 2 H^T L(Q)^T \frac{\partial f}{\partial x} \delta x$$
$$Q = f(x) \qquad |\hat{\omega}\theta| = 2 G(Q)^T \frac{\partial f}{\partial x} \delta x$$

$x \xrightarrow{f}$ Quaternion → axis-angle